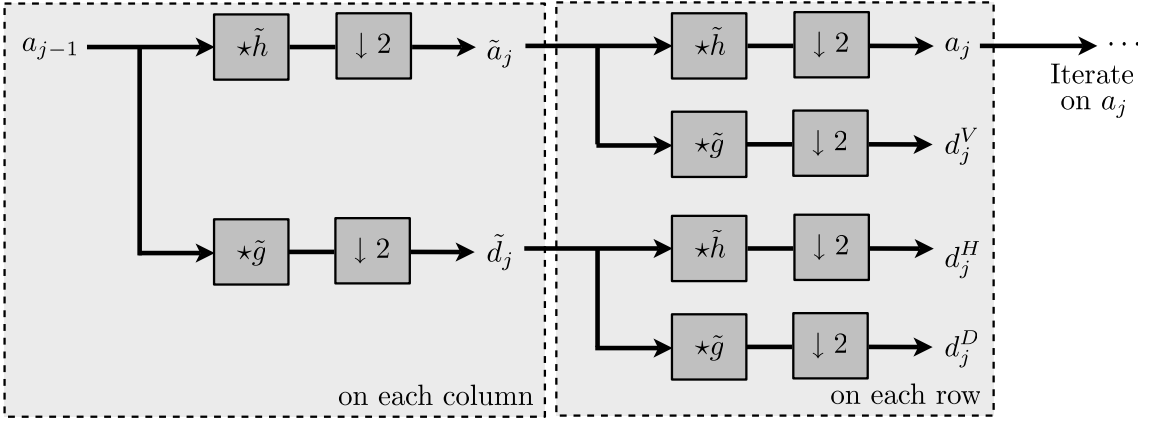
a_{j-1}
$\star\tilde{h}$
$\downarrow 2$
$\tilde{a}_j$
$\star\tilde{g}$
$\downarrow 2$
$\tilde{d}_j$
on each column
$\star\tilde{h}$
$\downarrow 2$
a_j
$\star\tilde{g}$
$\downarrow 2$
d_j^V
$\star\tilde{h}$
$\downarrow 2$
d_j^H
$\star\tilde{g}$
$\downarrow 2$
d_j^D
on each row
$\cdots$
Iterate on a_j